SMARTBRIDGE SKILL WALLET

Initial Project Planning Template

Date	04-07-2026
Team ID	
Project Name	Smart Agricultural Production Optimization Engine (OptiCrop)
Maximum Marks	5 Marks

Product Backlog, Sprint Schedule, and Estimation (4 Marks)

Sprint	Functional Requirement (Epic)	User Story Number	User Story / Task	Story Points	Priority	Team Members	Sprint Start Date	Sprint End Date (Planned)
Sprint-1	Data Collection & EDA	USN-1	As a developer, I can collect the crop recommendation dataset and load it using Pandas	2	High	Voleti Pavan Teja	01-07-2026	02-07-2026
Sprint-1	Data Collection & EDA	USN-2	As a developer, I can perform univariate, bivariate, and multivariate analysis to understand feature relationships	3	High	Voleti Pavan Teja	01-07-2026	02-07-2026
Sprint-1	Data Pre-processing	USN-3	As a developer, I can check for null values, review outliers, and extract seasonal crop groupings	2	High	Voleti Pavan Teja	01-07-2026	02-07-2026
Sprint-1	Data Pre-processing	USN-4	As a developer, I can split the dataset into training and test sets using stratified sampling	1	High	Voleti Pavan Teja	01-07-2026	02-07-2026
Sprint-2	Model Building	USN-5	As a developer, I can apply K-Means clustering to explore natural crop groupings	2	Medium	Voleti Pavan Teja	02-07-2026	03-07-2026
Sprint-2	Model Building	USN-6	As a developer, I can train a Logistic Regression model with scaled features to predict the suitable crop	3	High	Voleti Pavan Teja	02-07-2026	03-07-2026
Sprint-2	Model Building	USN-7	As a developer, I can evaluate the model using accuracy, confusion matrix, and classification report	2	High	Voleti Pavan Teja	02-07-2026	03-07-2026

Sprint-2	Model Building	USN-8	As a developer, I can save the trained model and scaler using Pickle for deployment	1	High	Voleti Pavan Teja	02-07-2026	03-07-2026
Sprint-3	Application Development	USN-9	As a farmer, I can view a Home page introducing OptiCrop and its purpose	1	Medium	Voleti Pavan Teja	03-07-2026	04-07-2026
Sprint-3	Application Development	USN-10	As a farmer, I can view an About page explaining who OptiCrop is built for	1	Low	Voleti Pavan Teja	03-07-2026	04-07-2026
Sprint-3	Application Development	USN-11	As a farmer, I can enter Nitrogen, Phosphorus, Potassium, temperature, humidity, pH, and rainfall values on the Find Your Crop page	3	High	Voleti Pavan Teja	03-07-2026	04-07-2026
Sprint-3	Application Development	USN-12	As a farmer, I can view the recommended crop on a result page after submitting my inputs	2	High	Voleti Pavan Teja	03-07-2026	04-07-2026
Sprint-4	Testing & Deployment	USN-13	As a developer, I can test the application locally with multiple known crop input profiles	2	High	Voleti Pavan Teja	04-07-2026	04-07-2026
Sprint-4	Testing & Deployment	USN-14	As a developer, I can push the project to GitHub for version control	1	Medium	Voleti Pavan Teja	04-07-2026	04-07-2026
Sprint-4	Testing & Deployment	USN-15	As a developer, I can deploy the application on Render so it is accessible via a public URL	2	High	Voleti Pavan Teja	04-07-2026	04-07-2026